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Performance Optimization in Zoho Creator

1. Aim

To demonstrate performance optimization techniques in a Zoho Creator application, particularly efficient record retrieval and workflow processing.

2. Introduction

Performance optimization in low-code applications means designing forms, workflows, and scripts so that they use data and processing only as far as needed. Although a low-code platform handles much of the underlying work, the way in which a workflow is written still influences how efficiently it runs.

Retrieving more records than required, or repeatedly processing records that are not relevant to the task, causes unnecessary data retrieval and processing. In a small application this may not be noticeable, but as the number of records and users grows it can reduce application efficiency. Efficient workflows in Zoho Creator therefore retrieve only the data they need and avoid unnecessary processing, which supports scalable and maintainable applications.

3. Application Used

The practical was performed in Zoho Creator 6 using the **Student Admission Enquiry Form** application. Deluge scripting was used to write the logic of a form workflow, *"Notify the student"*, that sends an email notification when an admission enquiry is submitted. The Zoho Creator account was on a trial plan, so some advanced features required an upgrade; no upgrade-only feature is claimed as enabled or configured in this record.

4. Performance Optimization Concept

The key optimization demonstrated is **criteria-based record fetching**. Instead of retrieving every record of a form, a criteria is given so that only the required record is returned. In the Student Admission workflow, this is written as:

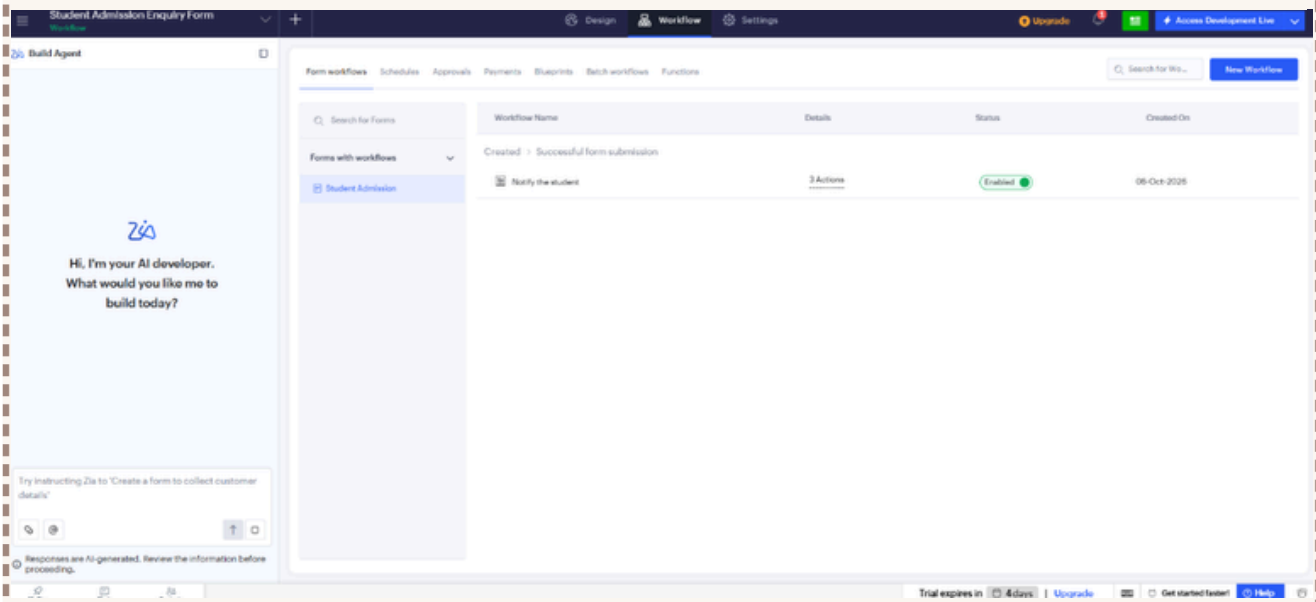
```
studentRecord = Student_Admission [ID == input.ID];
```

- The workflow identifies the current record using `input.ID`, which holds the ID of the record being processed.
- Only the relevant Student Admission record, the one whose ID matches `input.ID`, is retrieved.
- This avoids unnecessarily fetching and processing unrelated records.
- This improves workflow efficiency and reduces unnecessary data processing.

5. Procedure

1. Open the **Student Admission Enquiry Form** application in Zoho Creator in Edit mode.
2. Navigate to the **Workflow** section and open Form workflows. The form *Student Admission* is listed under Forms with workflows.
3. Observe the form workflow *"Notify the student"*. It is triggered on *Created Successful* form submission, has 3 actions, and has the status **Enabled** (Figure 1).
4. When a Student Admission form is submitted, the workflow identifies the current record using input.ID.
5. The required record is fetched using the criteria ID == input.ID, so that only this record is retrieved.
6. The workflow checks whether the email field of the record is blank.
7. If the email is available, the notification email is sent to the student. If the email is missing, an informational message is displayed.

Form Workflow of the Student Admission Enquiry Form Application



6. Deluge Code

The Deluge script used in the workflow is given below:

```
// Fetch only the specific record using criteria-based retrieval studentRecord =
Student_Admission [ID == input.ID]; if (studentRecord.email.isBlank()) { info "Email address
is missing for this enquiry"; } else { sendmail [ from : zoho.adminuserid to :
studentRecord.email subject : "Your Admission Enquiry has been Received" message : "<p>Dear
Student, </p><p>Your admission enquiry has been received. We will contact you shortly with
further details.</p><p>Thank you for your interest in our institution.</p>" ] }
```

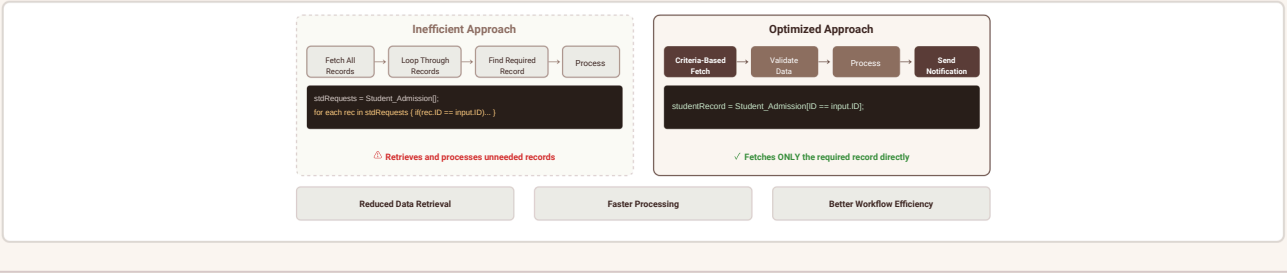
The first statement fetches the record by its ID. If the email field of that record is blank, an informational message is displayed. Otherwise, an email with the subject *"Your Admission Enquiry has been Received"* is sent to the student's email address.

7. Performance Analysis

The optimized approach is compared with a general inefficient approach below. The inefficient approach retrieves and processes records that are not needed, whereas the optimized approach reduces unnecessary record retrieval and processing.

Approach	Steps & Execution Pattern
Before (Inefficient Approach)	Fetch all records → Loop / iterate through records → Find required record → Process
Optimized (Used in Workflow)	Fetch the required record directly using ID==input.ID → Process directly

Figure 2: Inefficient and Optimized Approaches (Conceptual Diagram)



Note: No benchmark was performed in this practical, so no measured improvement in execution time is claimed. The comparison is based on the amount of data retrieved and processed.

8. Optimization Techniques

General practices that contribute to performance in Zoho Creator applications are detailed on the following page and summarized.

Figure 3: Performance Optimization Techniques (Conceptual Framework)

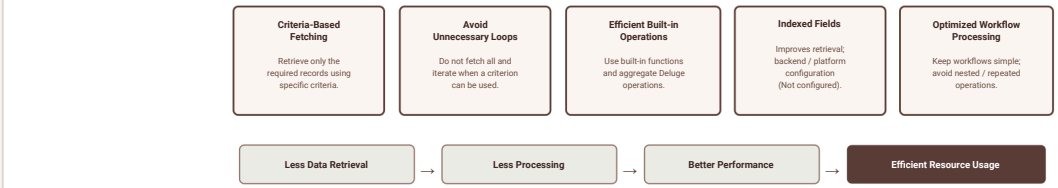
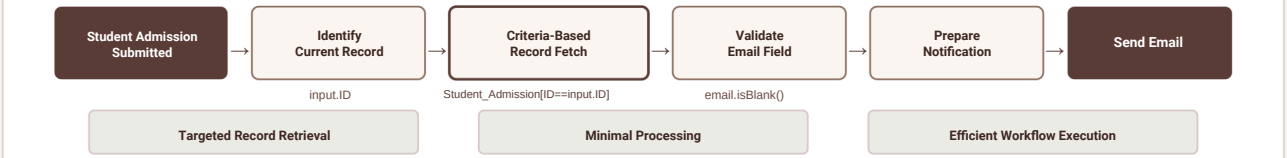


Figure 4: Optimized Student Admission Workflow Execution Flow



9. Conclusion

Performance optimization is important in low-code applications because efficiency depends not only on the platform but also on how workflows and scripts are written. In this practical, the Student Admission workflow used criteria-based record retrieval with the expression `Student_Admission [ID == input.ID]`, so that only the required record was retrieved and processed.

Targeted record retrieval improves efficiency by avoiding the retrieval of unrelated records and the iteration through them, and efficient workflow design reduces unnecessary processing. Practices such as criteria-based fetching, avoiding unnecessary loops, using built-in operations, and keeping workflows simple help applications to handle growing amounts of data. These practices contribute to scalable and maintainable Zoho Creator applications. No benchmark was performed, so the benefits are described in terms of reduced data retrieval and processing and not as measured execution times.